



Feasibility Study

Digital Certificate Verification System

Faculty of Computers and Information, Menoufia University | System Analysis & Design Project

1. Feasibility Assessment

Type	Current Situation / Considerations	Proposed Solution / Benefits	Feasibility Result
Technical	Certificates are verified manually (paper letters, phone calls); no central digital record; no real-time validation; easy to forge.	Web-based system with a database (MySQL/PostgreSQL); unique verification code and QR code per certificate; hash-based tamper detection; cloud hosting with automatic backups.	✓ Technically feasible
Operational	Verification requests are repetitive and time-consuming for Student Affairs staff; employers cannot verify certificates themselves; staff may resist changing the routine.	Instant self-service verification for employers; admin dashboard to add, update and revoke certificates; reduced staff workload; simple interface with short staff training.	✓ Operationally feasible
Economic	Cost of software development, hosting, training, maintenance and digitizing legacy records (see Cost-Benefit Analysis sheet).	Saves staff time, paper and printing; reduces forgery-related risk; lower admin costs and long-term efficiency gains.	✓ Cost-effective
Legal	Graduate personal data must be protected; certificate records must stay accurate and officially valid.	Role-based access control, encrypted data, minimal data shown on the public page, audit log; comply with data-protection regulations (e.g., Egypt Personal Data Protection Law No. 151 of 2020) and university policies.	✓ Legally feasible
Schedule	Manual processes delay verification and reporting.	Estimated timeline 3-6 months (requirements → design → development → testing → deployment).	✓ Achievable timeline

2. Cost-Benefit Analysis

All amounts are estimates in US dollars. Total costs exclude the existing manual-process cost (Administration), which is counted as a yearly saving.

Category	Description	Estimated Cost - Low (\$)	Estimated Cost - High (\$)	Benefits	Estimated Value (per year)
Software	In-house development (web app, verification page, admin dashboard) or purchase of a license	\$1,500	\$4,000	Automated verification reduces errors and manual handling	\$2,500
Hardware & Hosting	Cloud server/hosting, domain, SSL, backup storage	\$150	\$600	Instant online verification; far fewer verification requests handled by staff	\$600
Training	Training sessions for Student Affairs staff	\$300	\$800	Smooth transition and efficient use of the system	\$800
Maintenance	Annual updates, bug fixes, security patches, hosting renewal	\$700	\$1,200	Ensures system reliability and security; avoids downtime and fraud cases	\$1,500
Legacy Data Digitization	Entering previous graduates' certificate records into the system	\$600	\$1,500	Searchable records; faster retrieval of old certificates	\$700
Administration	Current manual process (paper letters, printing, courier, staff time) - existing yearly cost	-	-	Eliminated with the system → paperless verification	\$1,200
User Experience	Employers and graduates wait days for manual verification	-	-	Verification in seconds; improved satisfaction and trust	Intangible but high value
Total		\$3,250	\$8,100		\$7,300

Summary

Total Estimated Costs (first year)	\$3,250 - \$8,100
Total Estimated Annual Benefits/Savings	\$7,300
Payback Period	within 1 year
Conclusion	The Digital Certificate Verification System is economically feasible: the investment is recovered in 1 year, with clear long-term value.